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Short Stature Syndromes: Case Series from India

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Abstract

Syndromes causing short stature include Noonan syndrome (NS), Williams syndrome, and Silver–Russell syndrome (SRS). SRS is a primordial dwarfism with genetic heterogeneity. The SRS children present with prenatal growth retardation, neonatal hypoglycemia, feeding difficulties, physical asymmetry, with scoliosis and cardiac defect in some cases. The incidence is up to 1 in 100,000. Uniparental disomy, methylation abnormalities, and variants in some genes have been found underlying such phenotype. Growth hormone therapy has been used to improve the height gain in these patients. NS has genetic heterogeneity and most patients present with short stature with or without cardiac defect. Multiple genetic variants, mostly autosomal dominant, contribute to the phenotype. With the availability of next-generation sequencing, more and more genetic disorders causing short stature are being identified in different ethnic populations like Kabuki syndrome and Nance–Horan syndrome. Here, we present some cases of SRS and other additional syndromes with dysmorphism seen in past 5 years.

Keywords

- Noonan syndrome
- Kabuki syndrome
- Silver–Russell syndrome
- MLPA
- NGS

Introduction

Several genetic syndromes causing short stature are considered rare. Relatively common ones seen in the genetic clinic are Williams syndrome (WS, OMIM # 194050) and Noonan syndrome (NS). There are several causes of prenatal and postnatal growth retardation. These disorders are included under the term "primordial dwarfism."[1,2] In Silver–Russell syndrome (SRS), the head circumference is relatively preserved and additional findings include body asymmetry, hypoglycemia, and feeding difficulties.[1] Few of these patients also have a cardiac defect. Maternal uniparental disomy (UPD) of chromosome 16 or UPD(16)mat and UPD(7)mat have been described in some of the patients. Hypomethylation of chromosome 11p15 has been found in over 50% of the cases. Recently, mutations in *IGF2*, *CDKN1C,* and *PLAG1* have also been described in patients with such phenotype.[3] NS was first identified as one of the syndromes with Turner syndrome stigmata. However, with further advancements in diagnostic technologies, the genetic heterogeneity and variable presentations are being increasingly identified.[4,5] Certain syndromes with short stature can be X-linked, and these include Nance–Horan syndrome (NHS; OMIM # 302350) and faciogenital dysplasia (FGD; OMIM # 305400).[6,7] We report children with SRS and few other syndromes like Kabuki syndrome (KS) and FGD, seen in the genetic clinic of a tertiary care center.

Methods

This is a retrospective analysis of syndromes presenting with short stature seen in genetic clinic and genetic ward of a tertiary care center during a 5-year period from January 2016 to October 2020. Of total of 70 cases, 10 cases of WS and 38 cases of mucopolysaccharidoses were excluded from the detailed analysis (►**Fig. 1**). All patients underwent ultrasonography of the abdomen, cardiac evaluation, X-rays of the dorsolumbar spine and hands with wrist, along with thyroid profile as per the standard protocol in the genetic–metabolic

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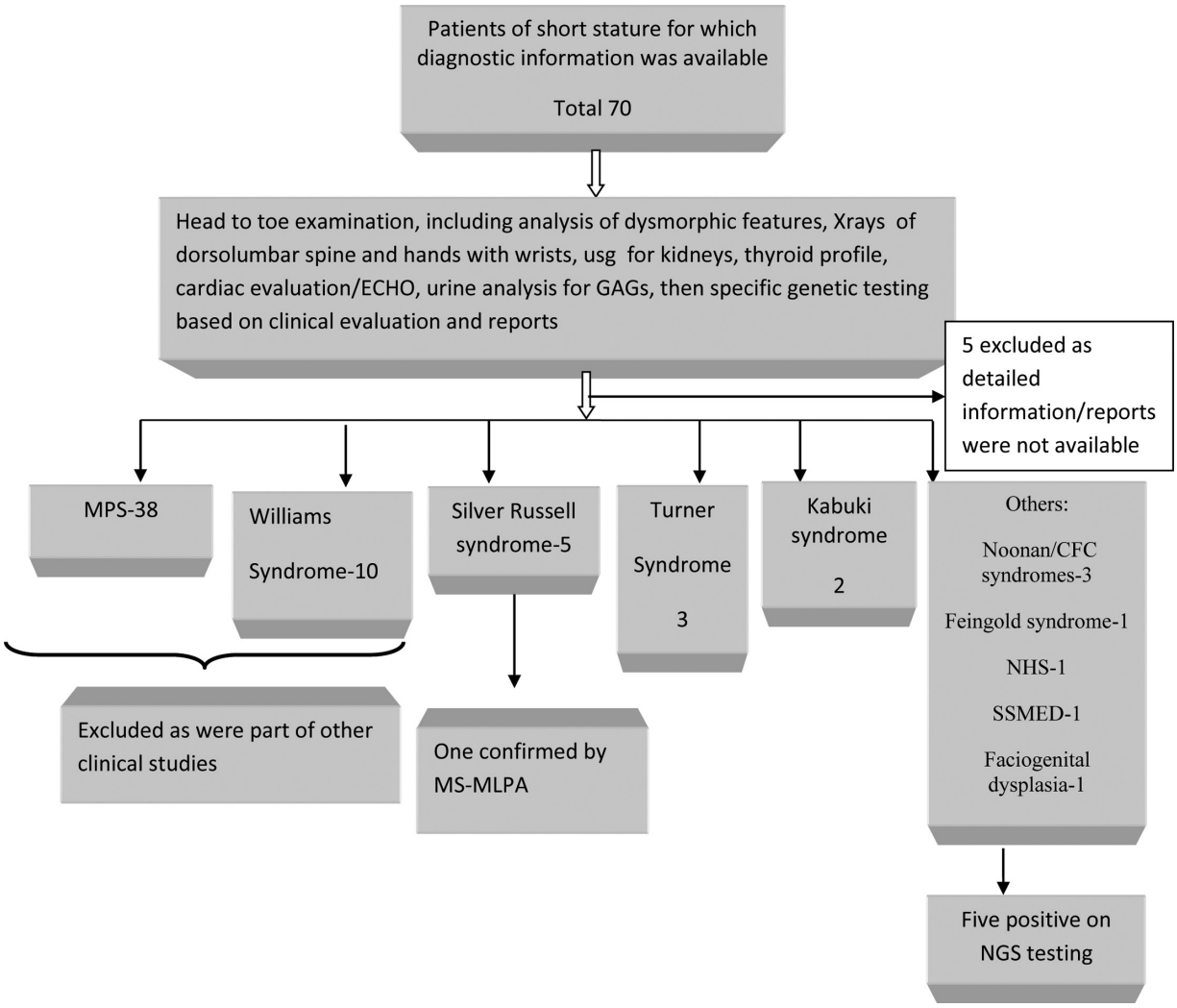


Fig. 1 Flow chart of patients with short stature seen in genetic clinic. ECHO, echocardiography; GAGS, glycosaminoglycans; MS-MLPA, methylation sensitive multiplex ligation dependent probe amplification; NGS, next-generation sequencing, USG, ultrasonography.

unit. Further investigations like karyotyping, methylation-sensitive multiplex ligation probe amplification (MS-MLPA), or next-generation sequencing (NGS) were done as per further requirement depending on the clinical diagnosis. Karyotyping was done for suspected Turner syndrome, especially females with unexplained short stature. FISH or chromosomal microarray was done in suspected WS. MS-MLPA was performed in those with prenatal and postnatal growth retardation where SRS was suspected. Subsequently, NGS was done; or if NS or KS was suspected from facial dysmorphic features/facial gestalt. Genetic testing was done with proper pretest counseling and informed consent. Clearance for publication was through the departmental review board. The data prioritization, filtering, and reporting on NGS were done as per standard American College of Medical Genetics (ACMG) guidelines.[8]

Results

Of 70 cases of short stature, there were 49 males and 21 females. The relatively common chromosomal syndromes seen were WS and Turner syndrome. The other syndromes and disorders included mucopolysaccharidosis that was the largest group, followed by SRS and NS. The details of types of syndromes seen are listed in ►**Fig. 1**. Some interesting cases are described below, including one SRS case (case 1), confirmed by MS-MLPA. The details of four other SRS patients (cases 2–5) are listed in ►**Table 1**, with Netchine–Harbison criteria.[9] Characteristic clinical diagnostic features were found in all cases.

Case 1

A 7-year-old female was brought with complaints of poor weight gain noticed by the parents compared with her peers. She was born at full term, with a birth weight of 2 kg, and had no perinatal problems. Her developmental milestones were normally achieved and she had another younger male child who was thriving well. Examination revealed weight of 14 kg (–3.62 z-score), height of 107.6 cm (–2.76 z-score), and head circumference of 48.5 cm (–2.0 z-score). The upper segment and lower segment ratio was 1:1. Her arm span was 98.3 cm. The head-to-toe examination showed a triangular face, with epicanthic folds, and low set ears, and downturned corners of mouth (►**Fig. 2**). There were few cafe-au-lait spots over the abdomen, and she had bilateral brachydactyly and

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Table 1 Clinical features of patients with Silver–Russell syndrome

Sr. no	Clinical criteria	Case 2	Case 3	Case 4	Case 5
1.	SGA (birth weight and/or birth length)	1.75 kg	1.00 kg	2.1 kg	1.7 kg
2.	Postnatal growth failure (SD)	−2.56	−4	−2.18	−3.92
3.	Relative macrocephaly at birth	NA	NA	NA	NA
4.	Prominent forehead	Yes	Yes	Yes	Yes
5.	Body asymmetry	Yes	Yes	Yes	Yes
6.	Feeding difficulties and/or low BMI	Yes	Yes	Yes	Yes
7.	Age at presentation	10 years, 2 M	1 year, 4 M	8 years, 5 M	5 years, 4 M
8.	Sex	Male	Male	Male	Male
9.	Additional clinical features	Triangular face, clinodactyly of fifth finger, retrognathia, relative macrocephaly at presentation (2 SD)	Triangular face, clinodactyly of fifth finger, syndactyly of toes, squeaky voice, occipital plagiocephaly, hypospadias, relative macrocephaly at presentation (≤ 2 SD)	Triangular face, clinodactyly of fifth finger, syndactyly of second and third toes, squeaky voice, occipital plagiocephaly	Triangular face, anteverted ears, retrognathia, overhanging columella, joint laxity, single ectopic kidney, cardiac defects, head circumference– (48.6 cm)

Abbreviations: BMI, body mass index; SD, standard deviation; SGA, small-for-gestational age.



Fig. 2 Silver–Russell syndrome (SRS) child showing facial asymmetry, hypertelorism, and flat facial profile.

clinodactyly. She also had partial syndactyly of second to third toes. The lower limb showed asymmetry with length of 53.8 cm on the left and 56.3cm on the right. She had delayed bone age on wrist X-rays corresponding to 5.5 years. Her thyroid profile was normal with T3/T4/TSH (thyroid-stimulating hormone) of 1.85/10.5/3.7 units. MS-MLPA showed loss of methylation of *H19DMR* region on chromosome 11p15 confirming SRS. Growth hormone (GH) stimulation study adequate response to stimulation. Her IGF1 level ws273 units ($n = 57$–316); and magnetic resonance imaging (MRI) brain did not reveal any pituitary abnormalities. The karyotype was 46, XX. On follow-up after 6 months of recombinant human GH therapy, her height was 111.4 cm and weight was 15 kg.

Case 6

A 9-year-old female was referred to genetic outpatient clinic in view of failure to thrive with dysmorphism. She was born out of nonconsanguineous marriage as preterm at 35 weeks gestation, born through normal vaginal delivery with birth weight of 2 kg and. Antenatally, there was history of polyhydramnios. At birth, she was found to have duodenal atresia for which she was operated at 6 weeks of age. Thereafter, she was asymptomatic except for poor gain in weight and height. The family history was unremarkable.

On examination, vital signs were stable, weight of child was 16 kg corresponding to –2.8 z-score, height was 114.6 cm (–4.56 z-score), head circumference was 46 cm (–5.6 z-score), body mass index was 12.4 (–1.8 z-score), and upper segment to lower segment ratio was 1.02:1. She had triangular face, prominent forehead, upward eye slant, epicanthic folds, and high arched palate with bifid uvula (►**Fig. 3**). Her hands and feet revealed brachydactyly, bilateral clinodactyly, bilateral syndactyly of fourth and fifth toes

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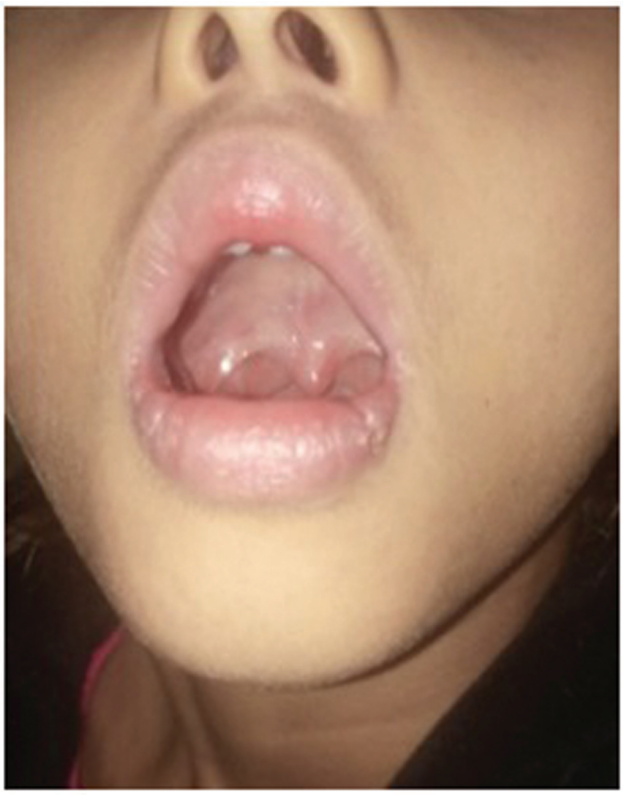


Fig. 3 Bifid uvula in child with Feingold syndrome.

(►**Fig. 4**). She also had flexion deformity of bilateral thumbs and an anal dimple. A clinical diagnosis of Feingold syndrome was made.

In routine investigations, complete blood counts, liver function test, renal function tests, and thyroid profile were normal. The bone age X-ray showed age corresponding to 7 years, chronological age being 9 years. Ultrasonography and echocardiography were done to rule out associated cardiac and renal malformations, which were normal.

Case 7

A 6-year-old male presented with generalized body swelling for 2 months, and cough for 1 week. He had previous history of decreased vision 3 years back and was operated for bilateral cataract, one eye at a time with a gap of 6 months.

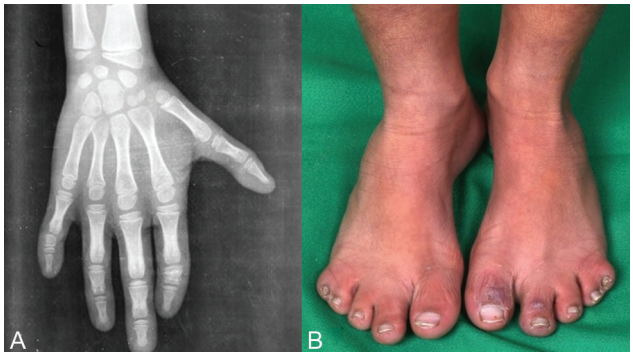


Fig. 4 X-ray of hand of Feingold syndrome child showing hypoplastic phalanges in second, fourth, and fifth fingers (**A**) and syndactyly of second to third and fourth to fifth toes (**B**).

He had thick scaly skin for 3 to 4 years and the skin biopsy was performed and it was suggestive of ichthyosis. Examination revealed mild coarse facies with pallor, generalized swelling over the body and dry hyperpigmented scales on limbs and trunk with relative sparing of the face (►**Fig. 5**). His weight was 10 kg (–4.67 z-score), height was 80 cm (–6.92 z-score), and head circumference of 46.5 cm (–4.7 z). He had cardiomegaly, liver of 3 cm below the right costal margin, mild ascites and decreased tone in the limbs. Echocardiography showed pericardial effusion. The thyroid profile was 0.19 ng/ml, 0.42 µg/dl, and 378.8 µIU/ml units for T3, T4, TSH, respectively. The thyroid peroxidase antibody level was 379.8 IU/ml and ultrasonography of thyroid showed hypoechoic and mildly bulky thyroid lobes. He received supportive therapy for respiratory distress, hydrocortisone, and thyroxine supplements. A pericardiocentesis was done to drain small amount of the pericardial fluid. The dysmorphic features were evident after the edema subsided and the features included hypertelorism, downslanting palpebral fissures strabismus, dysplastic, and low set anteverted ears and short neck. A final diagnosis of NS with hypothyroidism with ichthyosis was made on basis of NS diagnostic criteria.

Details of short stature cases 8 to 12 are listed in ►**Table 2**, in which variants related to phenotype were identified on NGS testing. Mother of case 9 had screwdriver incisors, suggested carrier state for NHS. She had a previous child with congenital cataract, who expired at 8½ months of age due to a diarrheal illness. The dysmorphic features of the two cases with KS are shown in ►**Fig. 6**. Two cases, one NS and one cardiofaciocutaneous syndrome with identified variants, have been described in an earlier report.

Discussion

There are several genetic causes of short stature, including Turner syndrome, WS, SRS, velocardiofacial syndrome, and NS. NS is associated mostly with pulmonic stenosis and in most cases the underlying cause is *PTPN11* mutations, though multiple genes have been implicated in causation of phenotype.[4,5,10] SRS has been described as a disorder of abnormal methylation and imprinting. In up to 10% cases UPD(16)mat or UPD(7)mat have been identified.[1,2] Nowadays, it is possible to identify hypomethylation of the *H19DMR* region by using MS-MLPA or NGS-based techniques in 50 to 60% cases.[9,11] In one of the cases in present report, we also found hypomethylation in the region. Based on clinical criteria also, one can diagnose SRS as elucidated in the Netchine–Harbison criteria as listed in ►**Table 1**.[1,9] Points 1 to 6 in ►**Table 1** are the clinical diagnostic criteria for SRS, which include four objective and two subjective criteria. The body asymmetry can be confirmed by measurement of limb length and limb girth in the patients using standard guidelines. In few cases, whole body on one side may be smaller, called hemihypotrophy. X-rays can help in diagnosis of hip dysplasia. History of neonatal hypoglycemia may also be found in SRS patients. The differential diagnosis includes 3M syndrome, Floating-Harbor syndrome, and SHORT syndrome; the first two also cause primordial dwarfism.[12]

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Fig. 5 Ichthyosis in child with Noonan syndrome and immune-mediated hypothyroidism.

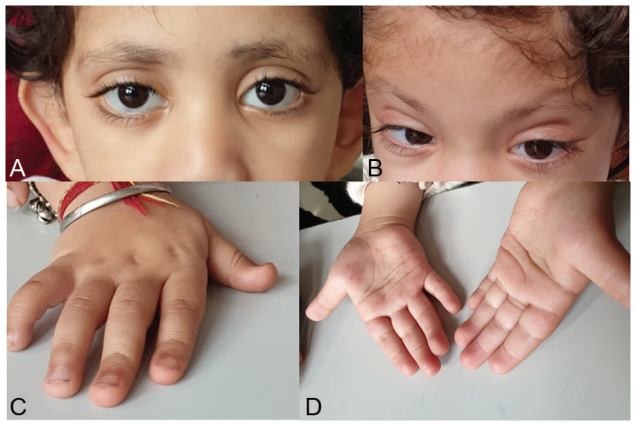


Fig. 6 Dysmorphic features in Kabuki syndrome showing blue sclera, anteverted ears, eversion of lateral portion of lower eyelids (**A**), sparse eyebrows (**A** and **B**), short hands (**C**), and persistent fingertip pads (**D**).

SHORT refers to constellation of findings in addition to **s**hort stature, including also **h**yperextensibility of joints /hernias, **o**cular depression, **R**ieger anomaly, and delay in **t**eething. In these syndromes, joint hypermobility or joint laxity is usually observed. Relative macrocephaly also occurs in 3M syndrome. MLPA not only helps in confirmation of diagnosis but also in cases of suspected WS, and MS-MLPA can diagnose Prader–Willi syndrome. Methylation disorders like SRS and Beckwith-Weidemann syndrome (BWS) are more likely to be seen in children born of assisted reproductive technologies, and MS-MLPA can help in diagnosing both.[11] Hypermethylation of *H19DMR* region occurs in BWS. MS-MLPA can also detect 11p15.5 duplications. In a recent report, use of NGS enabled identification of variants in three genes in different patients with SRS phenotype; notably *IGF2*, along with *CDKN1C* and *PLAG1*.[3]

Feingold syndrome is the most frequent single-gene cause of esophageal and duodenal atresia.[13] The presence of mild-to-moderate microcephaly and digital anomalies is highly suggestive of a diagnosis of Feingold syndrome. The constellation of features with duodenal atresia, failure to thrive, microcephaly, characteristic hand, and foot anomalies including clinodactyly, syndactyly between fourth and fifth toe, short toes, high arched palate, antenatal polyhydramnios are characteristic of Feingold syndrome. It is also called as microcephaly-oculo-digito-esophago-duodenal

Table 2 Clinical details of short stature patients—cases 8–10

Patient id	Age/sex	Clinical features	Genetic testing
Case 8	3 years/M	Poor growth, short stature, bilateral ptosis, widow's peak, short hands with clinodactyly in right hand, flat feet, no genital abnormality	NM_004463.2 (*FGD1*): Exon 9–18 del; 10514 bp hemizygous, identified on NGS Pathogenic X-linked recessive (XLR)
Case 9	7 years/M	Language delay and bilateral cataract (operated) Dysmorphism-coarse facies, anteverted ears, prominent nose Dental-diastema Behavioral abnormalities—hyperactive behavior, aggression present	NM_001291867.1 (*NHS*): c.666C > A: p.(Cys222*) hemizygous, XLR Pathogenic
Case 10	2 years/M	Failure to thrive, sparse scalp hair, triangular face, prominent forehead, straight eyebrows, deep set eyes, anteverted ears, squeaky voice, bilateral clinodactyly, second and third toe partial syndactyly of and loss of subcutaneous truncal adipose tissue	ENST00000511817.1 *XRCC4*:c.127T > C:p.(Trp43Arg); Exon 2, homozygous, pathogenic
Case 11	3 years/F	Microcephaly, prominent forehead, and large anteverted ears, intellectual disability, hyperactivity, without any seizures	CMA: 713 kb duplication on 10q11.22 NGS: Variant in exon 48: *KMT2D*:c.14890delC (p.Arg4964Valfs*30)
Case 12	5 years/F	Subtle seizures, short stature, microcephaly, dysmorphism-blue sclera, sparse eyebrows, short hands; persistent fingertip pads	*KMT2D* gene exon 39 variant, on NGS_NM_003482.3: c.11475_11478delACAG:p.(Gln3826Cysfs*3)

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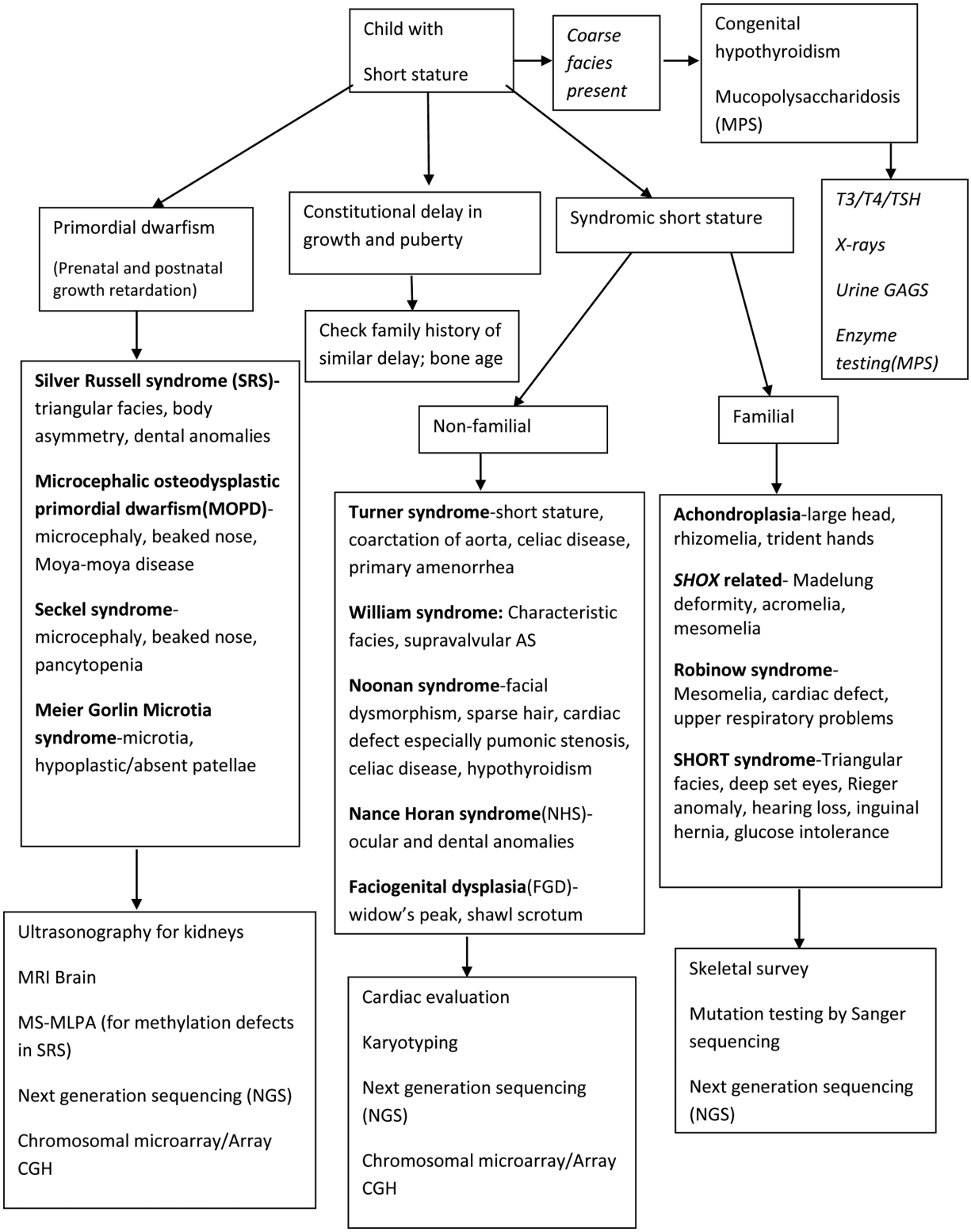


Fig. 7 Short stature: A clinical approach to diagnosis (Please note: spondylometaphyseal dysplasias and other rare skeletal dysplasias, and obesity syndromes are not included in the algorithm). CGH, comparative genomic hybridisation; MS-MLPA, methylation-sensitive multiplex ligation probe amplification.

syndrome. The patient in this report had short toes with syndactyly of fourth and fifth toes. The closest differential of Feingold syndrome is VACTERL (Vertebral defects, Anorectal malformations, Cardiac defects, Tracheo-esophageal fistula, Renal malformations, and Limb defects) and it is the most frequent diagnosis made in patients with multiple congenital anomalies involving esophageal, duodenal, or anal atresia.[14]

Faciogenital dysplasia or Aarskog syndrome causes short stature with dysmorphism, and clue to diagnosis is widow's peak with genital abnormality.[7,15] However, the child in present series did not have any genital abnormality like

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shawl scrotum. The variant found in the child in ►**Table 2**—*FGD1*: deletion of xon 9–18, has been described as likely pathogenic in an earlier report.[15] In NHS, there are mutations in *NHS* gene on short arm of X chromosome-Xp22.2-p22.1. Lens opacities are seen in 90% of heterozygous males—lens opacities involve posterior Y-sutures. The other ocular abnormalities are microphthalmia, microcornea, and strabismus; and most affected males have significant visual impairment. The NHS protein colocalizes with a tight junction protein in the cell membrane of epithelial cells and is involved in cataractogenesis.[6] A nonsense hemizygous variant was observed—*NHS*:c.666C > A:p.(Cys222*)—in the 7-year-old patient in present series (►**Table 2**). In a family with multiple affected members of an Indian family with severe ocular findings, another truncating mutation *NHS*: c.115C > T:p.(Gln39*) has been described earlier.[16] Postnatal growth failure with endocrine abnormalities is also seen in defects of DNA repair mechanism involving nonhomologous end joining complex. Pathogenic variants in the *XRCC4* gene cause short stature with microcephaly and endocrine dysfunction.[17,18]The clinical features include microcephaly, short stature, hypothyroidism, cryptorchidism, inguinal hernia, cardiomyopathy, and later onset ataxia. The homozygous variant in exon 2 described in the case 10—*XRCC4*:c.127T > C:p.(Trp43Arg)—has earlier been described in ClinVar as a pathogenic variant.

KS presents with intellectual disability and dysmorphism. Recently diagnostic criteria for KS have been developed.[19] It was first described in Japan, but subsequently patients from different countries have been described. Immune manifestations are seen in around 25% cases and include thrombocytopenia, and autoimmune hemolytic anemia.[20] *KMT2D* gene variants account for over 90% cases of this syndrome.[19] Both patients in present report showed heterozygous truncating variants leading to premature truncation of protein. Eversion of lateral portion of lower eyelids is a characteristic diagnostic feature of KS. The variant seen in first patient of KS is possibly novel variant, *KMT2D*:c.14890delC(p.Arg4964Valfs*30) and not described in ClinVar database. The variant identified in the second patient (NM_003482.3: c.11475_11478delACAG:p.[Gln3826Cysfs*3]) has been described earlier by Digilio et al 3 years earlier.[21] Approach to diagnosis in a child with short stature is elucidated with flow diagram in ►**Fig. 7**. Thus, availability of variety of genetic tests has enabled better identification and characterization of patients with short stature seen in the genetic clinic. NGS helps in diagnosis of most cases of short stature syndromes including NS, KS, and NHS. Parental testing or sibling testing may be indicated if a variant of unknown significance is identified in cases of heterozygous variants associated with autosomal dominant disorder. In X-linked disorder like NHS, the mother may be having minor manifestations that should be carefully looked for like screw-driver incisors as seen in mother of case 9 in present report.

Genes causing short stature include *PTPN11, LZTR1,* and *SOS1* leading to NS.[22] Short stature with scoliosis can be caused by *ACAN* mutations.[23] Mutations in *ACAN* are associated with early onset osteoarthritis. Missense variants in *ARID1B* have also been implicated in nonsyndromic short stature.[24] Apart from *KMT2D* that accounts for over 90% cases of KS, KS is also caused by *KDM6A* and *KDM6C* heterozygous or hemizygous variants.[25] The latter gene causes X-linked form of the disorder. In some cases of KS, severe brain malformations can occur like hypogenesis of corpus callosum, hydrocephalus, cerebellar vermis hypoplasia, and Dandy–Walker malformation. Accordingly, skeletal survey, cardiac evaluation, and MRI brain may be indicated in complete evaluation of some of the patients as illustrated in ►**Fig. 7**.

Conclusions

A careful clinical examination with emphasis on clues in the face, eyes, hairs, and teeth along with use of available clinical diagnostic criteria can enable better dysmorphic diagnosis in cases of children with short stature. An appropriate genetic testing, like MLPA or NGS, can confirm the diagnosis and facilitate better genetic counselling and management.

Data Availability Statement

Data for the NGS patients is available with the corresponding author and not given at present due to privacy or ethical restrictions.

Conflict of Interest

None declared.

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References

1 Wakeling EL, Brioude F, Lokulo-Sodipe O, et al. Diagnosis and management of Silver-Russell syndrome: first international consensus statement. Nat Rev Endocrinol 2017;13(02):105–124
2 Khetarpal P, Das S, Panigrahi I, Munshi A. Primordial dwarfism: overview of clinical and genetic aspects. Mol Genet Genomics 2016;291(01):1–15
3 Inoue T, Nakamura A, Iwahashi-Odano M, et al. Contribution of gene mutations to Silver-Russell syndrome phenotype: multigene sequencing analysis in 92 etiology-unknown patients. Clin Epigenetics 2020;12(01):86
4 van der Burgt I, Berends E, Lommen E, van Beersum S, Hamel B, Mariman E. Clinical and molecular studies in a large Dutch family with Noonan syndrome. Am J Med Genet 1994;53(02):187–191
5 Romano AA, Allanson JE, Dahlgren J, et al. Noonan syndrome: clinical features, diagnosis, and management guidelines. Pediatrics 2010;126(04):746–759
6 Sharma S, Ang SL, Shaw M, et al. Nance-Horan syndrome protein, NHS, associates with epithelial cell junctions. Hum Mol Genet 2006;15(12):1972–1983
7 Schwartz CE, Gillessen-Kaesbach G, May M, et al. Two novel mutations confirm FGD1 is responsible for the Aarskog syndrome. Eur J Hum Genet 2000;8(11):869–874
8 Richards S, Aziz N, Bale S, et al; ACMG Laboratory Quality Assurance Committee. Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the

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Association for Molecular Pathology. Genet Med 2015;17(05): 405–424
9 Azzi S, Salem J, Thibaud N, et al. A prospective study validating a clinical scoring system and demonstrating phenotypical-genotypical correlations in Silver-Russell syndrome. J Med Genet 2015; 52(07):446–453
10 Malaquias AC, Noronha RM, Souza TTO, et al. Impact of growth hormone therapy on adult height in patients with PTPN11 mutations related to Noonan syndrome. Horm Res Paediatr 2019;91(04):252–261
11 Priolo M, Sparago A, Mammì C, Cerrato F, Laganà C, Riccio A. MS-MLPA is a specific and sensitive technique for detecting all chromosome 11p15.5 imprinting defects of BWS and SRS in a single-tube experiment. Eur J Hum Genet 2008;16(05):565–571
12 Online Mendelian Inheritance in Man. OMIM. McKusick-Nathans Institute of Genetic Medicine, John Hopkins University (Baltimore, MD). Accessed February 17, 2021 at: https://omim.org/
13 Celli J, van Bokhoven H, Brunner HG. Feingold syndrome: clinical review and genetic mapping. Am J Med Genet A 2003;122A(04): 294–300
14 Shaw-Smith C. Oesophageal atresia, tracheo-oesophageal fistula, and the VACTERL association: review of genetics and epidemiology. J Med Genet 2006;43(07):545–554
15 Bedoyan JK, Friez MJ, DuPont B, Ahmad A. First case of deletion of the faciogenital dysplasia 1 (FGD1) gene in a patient with Aarskog-Scott syndrome. Eur J Med Genet 2009;52(04):262–264
16 Ramprasad VL, Thool A, Murugan S, et al. Truncating mutation in the NHS gene: phenotypic heterogeneity of Nance-Horan syndrome in an asian Indian family. Invest Ophthalmol Vis Sci 2005; 46(01):17–23
17 Shaheen R, Faqeih E, Ansari S, et al. Genomic analysis of primordial dwarfism reveals novel disease genes. Genome Res 2014;24 (02):291–299
18 Guo C, Nakazawa Y, Woodbine L, et al. XRCC4 deficiency in human subjects causes a marked neurological phenotype but no overt immunodeficiency. J Allergy Clin Immunol 2015;136(04): 1007–1017
19 Adam MP, Banka S, Bjornsson HT, et al; Kabuki Syndrome Medical Advisory Board. Kabuki syndrome: international consensus diagnostic criteria. J Med Genet 2019;56(02):89–95
20 Margot H, Boursier G, Duflos C, et al. Immunopathological manifestations in Kabuki syndrome: a registry study of 177 individuals. Genet Med 2020;22(01):181–188
21 Digilio MC, Gnazzo M, Lepri F, et al. Congenital heart defects in molecularly proven Kabuki syndrome patients. Am J Med Genet A 2017;173(11):2912–2922
22 Kumari D, Chaudhary D, Panigrahi I, Rohit MK. Genetic defects in children with cardiac anomalies/malformations. Noonan and CFC syndromes. J Pediatr Genet 2020. DOI: 10.1055/s-0040-1721441
23 Panigrahi I, Angurana SK, Varma H, et al. Phenotypic heterogeneity of kyphoscoliosis with vertebral and rib defects: a case series. Clin Dysmorphol 2019;28(03):103–113
24 Yu Y, Yao R, Wang L, et al. De novo mutations in ARID1B associated with both syndromic and non-syndromic short stature. BMC Genomics 2015;16(01):701
25 Bögershausen N, Gatinois V, Riehmer V, et al. Mutation update for Kabuki syndrome genes KMT2D and KDM6A and further delineation of X-linked Kabuki syndrome subtype 2. Hum Mutat 2016; 37(09):847–864

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